

① Determinanti o matrice A are satisfice

$$A = \begin{pmatrix} 11 & 10 \\ 5 & 6 \end{pmatrix}$$

$$B = \begin{pmatrix} 11 & 10 \\ 5 & 6 \end{pmatrix} \quad B \text{ diagonalizabil?}$$

$$\det(B - \lambda I) = 0 \Rightarrow \begin{vmatrix} 11-\lambda & 10 \\ 5 & 6-\lambda \end{vmatrix} = 0 \xrightarrow{L_1 - L_2} \begin{vmatrix} 6-\lambda & 4+\lambda \\ 5 & 6-\lambda \end{vmatrix} = 0$$

$$(6-\lambda)^2 - 5(4+\lambda) = 0 \Rightarrow 36 - 12\lambda + \lambda^2 - 20 - 5\lambda = 0$$

$$\lambda^2 - 17\lambda + 16 = 0 \Rightarrow \lambda^2 - 16\lambda - \lambda + 16 = 0 \Rightarrow$$

$$\Rightarrow \lambda(\lambda - 16) - (\lambda - 16) = 0 \Rightarrow (\lambda - 16)(\lambda - 1) = 0 \begin{cases} \lambda_1 = 16 \\ \lambda_2 = 1 \end{cases}$$

$$a(\lambda_1) = 1$$

$$a(\lambda_2) = 1$$

$$Bv = \lambda_1 v \Rightarrow \begin{cases} 11x + 10y = 16x \\ 5x + 6y = 16y \end{cases} \Rightarrow \begin{cases} 10y = 5x \\ 5x = 10y \end{cases} \Rightarrow x = 2y$$

$$V_{\lambda_1} = \text{Sp} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\} \quad g(\lambda_1) = 1$$

$$Bv = \lambda_2 v \Rightarrow \begin{cases} 11x + 10y = x \\ 5x + 6y = y \end{cases} \Rightarrow \begin{cases} 10x + 10y = 0 \\ 5x + 5y = 0 \end{cases} \Rightarrow x = -y$$

$$V_{\lambda_2} = \text{Sp} \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\} \quad g(\lambda_2) = 1$$

$$a(\lambda_1) = g(\lambda_1)$$

$$a(\lambda_2) = g(\lambda_2)$$

$$\Rightarrow B \text{ diagonalizabil} \Rightarrow B = P \cdot D \cdot P^{-1}$$

$$\Rightarrow D = \begin{pmatrix} 16 & 0 \\ 0 & 1 \end{pmatrix}; P = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$$

$$\left(\begin{array}{cc|cc} 2 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right) \xrightarrow{L_1 - L_2} \left(\begin{array}{cc|cc} 1 & -2 & 1 & -1 \\ 1 & 1 & 0 & 1 \end{array} \right) \xrightarrow{L_2 - L_1} \left(\begin{array}{cc|cc} 1 & -2 & 1 & -1 \\ 0 & 3 & -1 & 2 \end{array} \right)$$

$$\xrightarrow{L_1 + \frac{2}{3}L_2} \begin{pmatrix} 1 & 0 & 1/3 & 1/3 \\ 0 & 3 & -1 & 2 \end{pmatrix} \xrightarrow{L_2/3} \begin{pmatrix} 1 & 0 & 1/3 & 1/3 \\ 0 & 3 & -1/3 & 2/3 \end{pmatrix}$$

$$\text{Dare } A^4 = B = P \cdot D \cdot P^{-1} = P \cdot \begin{pmatrix} \lambda_1^4 & 0 \\ 0 & \lambda_2^4 \end{pmatrix} \cdot P^{-1} \Rightarrow P^{-1} = \begin{pmatrix} 1/3 & 1/3 \\ -1/3 & 2/3 \end{pmatrix}$$

$$\Rightarrow A = P \cdot D \cdot P^{-1} = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1/3 & 1/3 \\ -1/3 & 2/3 \end{pmatrix}$$

$$A = \begin{pmatrix} 4 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1/3 & 1/3 \\ -1/3 & 2/3 \end{pmatrix} = \begin{pmatrix} 5/3 & 2/3 \\ 1/3 & 4/3 \end{pmatrix}$$

② Determinați termenul general al șirului care verifică $x_{n+2} = 5x_{n+1} - 6x_n$

$$x_0 = 1$$

$$x_1 = 5$$

$$\begin{pmatrix} x_{n+2} \\ x_{n+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 5 & -6 \\ 1 & 0 \end{pmatrix}}_A \begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix} = A \begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix} = A^2 \begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix} = \dots = A^{n+1} \begin{pmatrix} x_1 \\ x_0 \end{pmatrix}$$

$$\det(A - \lambda I) = 0 \Rightarrow \begin{vmatrix} 5-\lambda & -6 \\ 1 & -\lambda \end{vmatrix} = 0 \Rightarrow -\lambda(5-\lambda) + 6 = 0$$

$$\Rightarrow -5\lambda + \lambda^2 + 6 = 0$$

$$\lambda^2 - 2\lambda - 3\lambda + 6 = 0 \Rightarrow \lambda(\lambda - 2) - 3(\lambda - 2) = 0$$

$$(\lambda - 2)(\lambda - 3) = 0 \Rightarrow \begin{cases} \lambda_1 = 2 \\ \lambda_2 = 3 \end{cases}$$

$$Av = \lambda_1 v \Rightarrow \begin{cases} 5a - 6b = 2a \\ a = 2b \end{cases} \Rightarrow \begin{cases} 3a = 6b \\ a = 2b \end{cases} \Rightarrow a = 2b$$

$$V_{\lambda_1} = \text{Sp} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\} \Rightarrow g(\lambda_1) = 1$$

$$Av = \lambda_2 v \Rightarrow \begin{cases} 5a - 6b = 3a \\ a = 3b \end{cases} \Rightarrow \begin{cases} 2a = 6b \\ a = 3b \end{cases} \Rightarrow a = 3b$$

$$V_{\lambda_2} = \text{Sp}\left\{\begin{pmatrix} 3 \\ 1 \end{pmatrix}\right\} \Rightarrow g(\lambda_2) = 1$$

$$a(\lambda_i) = g(\lambda_i) \Rightarrow A \text{ diagonalizabilă} \rightarrow \Delta = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

$$P = \begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix}$$

$$\left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right) \xrightarrow{L_1 - L_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & -1 \\ 1 & 1 & 0 & 1 \end{array}\right) \xrightarrow{L_2 - L_1} \left(\begin{array}{cc|cc} 1 & 2 & 1 & -1 \\ 0 & -1 & -1 & 2 \end{array}\right)$$

$$\xrightarrow{L_1 + 2L_2} \left(\begin{array}{cc|cc} 1 & 0 & -1 & 3 \\ 0 & -1 & -1 & 2 \end{array}\right) \xrightarrow{L_2(-1)} \left(\begin{array}{cc|cc} 1 & 0 & -1 & 3 \\ 0 & 1 & 1 & -2 \end{array}\right) \Rightarrow P^{-1} = \begin{pmatrix} -1 & 3 \\ 1 & -2 \end{pmatrix}$$

$$A^{u+1} = P \cdot \Delta^{u+1} \cdot P^{-1} = \begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2^{u+1} & 0 \\ 0 & 3^{u+1} \end{pmatrix} \begin{pmatrix} -1 & 3 \\ 1 & -2 \end{pmatrix}$$

$$A^{u+1} = \begin{pmatrix} 2^{u+2} & 3^{u+2} \\ 2^{u+1} & 3^{u+1} \end{pmatrix} \begin{pmatrix} -1 & 3 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 3^{u+2} - 2^{u+2} & 3 \cdot 2^{u+2} - 2 \cdot 3^{u+2} \\ 3^{u+1} - 2^{u+1} & 3 \cdot 2^{u+1} - 2 \cdot 3^{u+1} \end{pmatrix}$$

$$\begin{pmatrix} x_{u+2} \\ x_{u+1} \end{pmatrix} = \begin{pmatrix} 3^{u+2} - 2^{u+2} & 3 \cdot 2^{u+2} - 2 \cdot 3^{u+2} \\ 3^{u+1} - 2^{u+1} & 3 \cdot 2^{u+1} - 2 \cdot 3^{u+1} \end{pmatrix} \begin{pmatrix} 5 \\ 1 \end{pmatrix}$$

$$x_{u+2} = 5(3^{u+2} - 2^{u+2}) + 3 \cdot 2^{u+2} - 2 \cdot 3^{u+2}$$

$$x_u = 3 \cdot 3^u - 2 \cdot 2^u = 3^{u+1} - 2^{u+1}$$

③ Verificati dacă matricea A este diagonalizabilă. În caz afirmativ diagonalizați-o. În caz contrar, jordanizați-o.

$$A = \begin{pmatrix} 5 & -1 \\ 1 & 3 \end{pmatrix} \quad \det(A - \lambda I) = 0 \Rightarrow \begin{vmatrix} 5-\lambda & -1 \\ 1 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (5-\lambda)(3-\lambda) + 1 = 0 \Rightarrow (5-5\lambda-3\lambda+\lambda^2+1) = 0 \Rightarrow$$

$$\Rightarrow \lambda^2 - 8\lambda + 16 = 0 \Rightarrow (\lambda - 4)^2 = 0 \Rightarrow \lambda_1 = 4 \Rightarrow a(\lambda) = 2$$

$$\boxed{\lambda = 4} \begin{cases} 5x - y = 4x \\ x + 3y = 4y \end{cases} \Rightarrow \begin{cases} x = y \\ x = y \end{cases} \Rightarrow V_{\lambda_1} = \text{Sp}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\} \Rightarrow g(\lambda) = 1$$

$$\Rightarrow a(\lambda_1) \neq a(\lambda_2) \Rightarrow A \text{ nu e diagonalizabilă}$$

$$\Rightarrow A \text{ jordanizabilă} \Rightarrow A = P \cdot J \cdot P^{-1}$$

$$J = \begin{pmatrix} \lambda_1 & 1 \\ 0 & \lambda_2 \end{pmatrix}$$

$$P = (v | w) \text{ unde } v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \text{ iar } (A - \lambda J)w = v \Rightarrow$$

$$\Rightarrow \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow \begin{cases} x - y = 1 \\ x - y = 1 \end{cases} \Rightarrow x = y + 1$$

$$\Rightarrow w = \begin{pmatrix} \alpha + 1 \\ \alpha \end{pmatrix}, \alpha \in \mathbb{R} \Rightarrow w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\text{Prin urmare, } J = \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix} \text{ și } P = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & : & 1 & 0 \\ 1 & 0 & : & 0 & 1 \end{pmatrix} \xrightarrow{L_2 - L_1} \begin{pmatrix} 1 & 1 & : & 1 & 0 \\ 0 & -1 & : & -1 & 1 \end{pmatrix} \xrightarrow{L_1 + L_2} \begin{pmatrix} 1 & 0 & : & 0 & 1 \\ 0 & -1 & : & -1 & 1 \end{pmatrix}$$

$$\xrightarrow{L_2(-1)} \begin{pmatrix} 1 & 0 & : & 0 & 1 \\ 0 & 1 & : & 1 & -1 \end{pmatrix} \Rightarrow P^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$$

$$A = P \cdot J \cdot P^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$$

$$A = \begin{pmatrix} 5 & -1 \\ 1 & 3 \end{pmatrix}$$

④ Diagonalizati matricea $A = \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix}$

$$\lambda = \pm i$$

$$\lambda = \alpha \pm i\beta \Rightarrow \begin{cases} \alpha = 0 \\ \beta = 1 \end{cases}$$

$$\lambda = i \quad (\text{se iau valoarea complex pozitivă})$$

$$Av = \lambda v \Rightarrow \begin{pmatrix} 1-i & 1 \\ -2 & -1-i \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} (1-i)x + y = 0 \\ -2x - (1+i)y = 0 \end{cases} \Rightarrow y = (i-1)x$$

$$\begin{aligned} x &= \alpha, \alpha \in \mathbb{C} \Rightarrow v = \begin{pmatrix} \alpha \\ (i-1)\alpha \end{pmatrix} \\ y &= (i-1)\alpha \\ \Rightarrow v &= \begin{pmatrix} 1 \\ i-1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} i \end{aligned}$$

$$V_\lambda = \text{Sp} \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \longrightarrow P = (\text{Re } v \mid \text{Im } v)$$

$$A = P \cdot Q \cdot P^{-1}$$

$$P = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

$$Q = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad \begin{matrix} \alpha=0 \\ \beta=1 \end{matrix}$$

$$\left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 1 \end{array} \right) \xrightarrow{L_2 + L_1} \left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{array} \right) \Rightarrow P^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$$\begin{aligned} P \cdot Q \cdot P^{-1} &= \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix} = A \end{aligned}$$

⑤ Calculate A^n . $A = \begin{pmatrix} -4 & 1 \\ -1 & -2 \end{pmatrix}$

$$\begin{aligned} \det(A - \lambda I) &= 0 \Rightarrow \begin{vmatrix} -4-\lambda & 1 \\ -1 & -2-\lambda \end{vmatrix} = 0 \Rightarrow (4+\lambda)(2+\lambda) + 1 = 0 \\ &\Rightarrow 8 + 4\lambda + 2\lambda + \lambda^2 + 1 = 0 \\ &\Rightarrow \lambda^2 + 6\lambda + 9 = 0 \Rightarrow (\lambda+3)^2 = 0 \\ &\lambda = -3 \quad a(\lambda) = 2 \end{aligned}$$

$$\begin{cases} -4x + y = -3x \\ -x - 2y = -3x \end{cases} \Rightarrow \begin{cases} y = x \\ -2y = -2x \end{cases} \Rightarrow x = y \Rightarrow V_\lambda = \text{Sp} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\} \Rightarrow g(\lambda) = 1$$

$$\Rightarrow a(\lambda) \neq g(\lambda) \Rightarrow A \text{ jordanizable}$$

$$A = P \cdot J \cdot P^{-1} \quad \text{unde } P = (v \mid w)$$

$$v = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{care } (A - \lambda J)w = v \Rightarrow \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow -x + y = 1 \Rightarrow w = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ pt. } x = 0 \Rightarrow P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$$\left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right) \xrightarrow{L_2 - L_1} \left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{array} \right) \Rightarrow P^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

$$J = \begin{pmatrix} -3 & 1 \\ 0 & -3 \end{pmatrix} \rightarrow J^n = \begin{pmatrix} (-3)^n & n(-3)^{n-1} \\ 0 & (-3)^n \end{pmatrix}$$

$$J^n = \begin{pmatrix} \lambda^n & n\lambda^{n-1} \\ 0 & \lambda^n \end{pmatrix}$$

$$A^n = P \cdot J^n \cdot P^{-1}$$

$$A^n = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} (-3)^n & n(-3)^{n-1} \\ 0 & (-3)^n \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \Rightarrow$$

$$A^n = \begin{pmatrix} (-3)^n & n(-3)^{n-1} \\ (-3)^n & (n-3)(-3)^{n-1} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

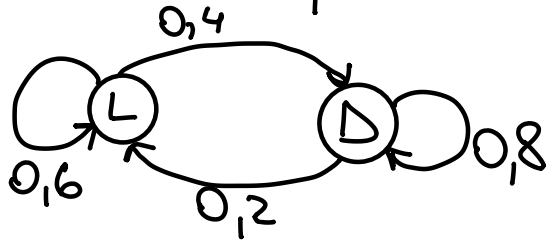
$$A^n = \begin{pmatrix} -(3+n)(-3)^{n-1} & n(-3)^{n-1} \\ -n(-3)^{n-1} & (n-3)(-3)^{n-1} \end{pmatrix}$$

$$A = P \cdot D \cdot P^{-1}$$

$$e^A = P e^D P^{-1}$$

$$e^D = \begin{pmatrix} e^{\lambda_1} & & \dots \\ & e^{\lambda_2} & \dots \\ & & \ddots \end{pmatrix}$$

6) Într-un campus universitar studenții joacă zilnic DOTA sau LOL. Sunt 20% șanse ca un jucător de DOTA să joace mâine LOL și 40% șanse ca un jucător de LOL să joace DOTA. Care este probabilitatea ca un jucător de LOL să joace LOL și peste 100 de zile.



x_1 - stud. care joacă DOTA

x_2 - stud. care joacă LOL

$$x_1(t+1) = 0.8x_1(t) + 0.4x_2(t)$$

$$x_2(t+1) = 0.2x_1(t) + 0.6x_2(t)$$

$$x(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$$

$$A = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix}$$

$$x(t) = A \cdot x(t-1)$$

$$x(t+100) = A \cdot x(t+99) = A^{100} \cdot x(t)$$

$$\det(A - \lambda I) = 0 \Rightarrow \begin{vmatrix} 0.8 - \lambda & 0.4 \\ 0.2 & 0.6 - \lambda \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} 1 - \lambda & 1 - \lambda \\ 0.2 & 0.6 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (1 - \lambda) \begin{vmatrix} 1 & 1 \\ 0.2 & 0.6 - \lambda \end{vmatrix} = 0 \Rightarrow (1 - \lambda)(0.6 - \lambda - 0.2) = 0$$

$$\Rightarrow (1 - \lambda)(0.4 - \lambda) = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = 0.4$$

$$\boxed{\lambda_1 = 1} \Rightarrow \begin{cases} 8x + 4y = 10x \\ 2x + 6y = 10y \end{cases} \Rightarrow \begin{cases} 4y = 2x \\ 2x = 4y \end{cases} \Rightarrow x = 2y$$

$$V_{\lambda_1} = \text{Sp} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$$

$$\boxed{\lambda_2 = 0,4} \Rightarrow \begin{cases} 8x + 4y = 4x \\ 2x + 6y = 4y \end{cases} \Rightarrow \begin{cases} 4x + 4y = 0 \\ 2x + 2y = 0 \end{cases} \Rightarrow x = -y \Rightarrow$$

$$\Rightarrow V_{\lambda_2} = \text{Sp} \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

$$A = P \cdot D \cdot P^{-1} \quad P = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \quad P^{-1} = \frac{1}{3} \begin{pmatrix} -1 & -1 \\ -1 & 2 \end{pmatrix}$$

$$D = \begin{pmatrix} 1 & 0 \\ 0 & 0,4 \end{pmatrix} \quad D^{100} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A^{100} = P \cdot D^{100} \cdot P^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \frac{1}{3} \begin{pmatrix} -1 & -1 \\ -1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix} \frac{1}{3} \begin{pmatrix} -1 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$$

$$A^{100} = \begin{pmatrix} 2/3 & 2/3 \\ 1/3 & 1/3 \end{pmatrix}$$

$\swarrow$ DOTA $\swarrow$ LOL
 $\leftarrow$ DOTA $\leftarrow$ LOL

$$\text{Prob.} = 1/3$$